

REVIEW

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Parkinson's disease: pathogenesis and therapeutic strategies

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Abstract

Parkinson's disease (PD) is a neurodegenerative disorder primarily characterized by motor impairments such as bradykinesia, tremor, and rigidity. Its neuropathological hallmarks include the progressive loss of dopaminergic neurons in the substantia nigra pars compacta (SNc) and the aggregation of α -synuclein (α -syn) into Lewy bodies (LBs), which gradually spread to other central nervous system regions and peripheral tissues. The etiology of PD is multifactorial, involving a complex interplay of genetic predisposition, aging, environmental exposures, and lifestyle factors. Disease pathogenesis is predominantly driven by the synergistic action of core pathological mechanisms, including α -syn aggregation, oxidative stress (OS), mitochondrial dysfunction, and neuroinflammation. In recent years, the role of peripheral-central communication pathways in disease initiation and propagation has garnered significant attention. To date, pharmacotherapy remains the mainstay for improving the quality of life in PD patients. The most commonly used clinical agents primarily target the replenishment of depleted dopamine in the brain. However, these medications only alleviate symptoms and do not slow disease progression. This inherent limitation underscores the urgent need for more effective therapeutic strategies. This review aims to systematically outline the network of PD pathogenesis and the evolution of its treatment strategies, with a particular emphasis on a holistic perspective from etiology to therapy. It critically evaluates current bottlenecks in drug treatment and provides an in-depth analysis of potential candidate drugs for PD, as well as the latest advances in α -syn-targeted immunotherapies, iPSC-based regenerative therapies, and gene therapies. Building on this foundation, we further argue that the future of PD management must shift towards integrated, multi-target, and personalized therapeutic strategies to overcome existing efficacy barriers.

Keywords Parkinson's disease, α -synuclein, Pathogenesis, Medication, Treatment progress

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Introduction

Parkinson's disease is the second most common neurodegenerative disorder after Alzheimer's disease. The average age of onset is around 60 years. With the aging of the global population, the incidence of PD is gradually increasing worldwide. Currently, approximately 1%–2% of people aged over 60 are affected by this disease [1]. The number of PD patients is projected to rise from 7 million in 2015 to 13 million by 2040 [2]. The pathological hallmarks of PD include the misfolding and aggregation of α -syn into LBs, leading to the degeneration of dopaminergic neurons in the substantia nigra pars compacta and the depletion of striatal dopamine. Its clinical manifestations are primarily motor dysfunctions and various non-motor symptoms [3]. As the disease progresses, all clinical symptoms worsen to varying degrees, imposing a significant burden on families and society [4].

As the precise pathological mechanisms of PD remain incompletely understood, current treatments mainly rely on pharmacological modulation of neurotransmitters in the brain. Dopamine replacement therapies such as levodopa (L-DOPA) remain the mainstream clinical treatment for PD [5]. However, these therapeutic approaches are primarily limited to symptom control and cannot delay or halt disease progression. Therefore, exploring novel therapeutic strategies for PD in the context of unclear etiology and pathogenesis presents a significant challenge for future anti-PD drug development.

In recent years, important advancements have been made in PD research. On the mechanistic level [6], investigations have expanded beyond traditional core pathways such as OS, mitochondrial dysfunction, and neuroinflammation to encompass complex networks involving multi-organ system interactions like the gut-brain axis and spleen-brain axis. This has revealed new hypotheses suggesting that α -syn may propagate from the periphery to the central nervous system along neural pathways [7]. On the therapeutic front, alongside novel drug delivery strategies like levodopa inhalation powder and continuous subcutaneous infusions [8], disease-modifying therapies targeting specific molecules such as α -syn and c-Abl, as well as cutting-edge approaches like immunotherapy, stem cell therapy, and gene therapy, have shown considerable potential. Against this backdrop, this review aims to provide a comprehensive and in-depth synthesis of the pathogenesis and therapeutic strategies for PD, hoping to offer insights for future research into PD mechanisms and the development of treatments and interventions.

The etiology of Parkinson's disease

The etiology of PD remains unclear and may involve factors such as genetics, environment, gender, age, and lifestyle [9]. With advancing age, the incidence of PD gradually increases, and aging constitutes the most significant risk factor for the disease [10]. Compared with females, males exhibit a higher incidence of PD [11]. Additionally, studies have found that gut microbiota plays an important role in regulating motor dysfunction [12]. Metabolites produced by gut microbiota may influence brain neurons, and PD patients often experience gastrointestinal dysfunction. Therefore, alterations in the gut microbiota may represent a risk factor for the development of PD.

Genetic factor

Genetic factors represent a common cause of PD, with a substantial proportion of PD cases influenced by genetic components [13]. In 1997, the first PD-associated gene, SNCA, was identified in an Italian family. It is an autosomal dominant gene located on human chromosome 4q21-23, containing six exons and encoding a 140-amino acid protein. SNCA is involved in encoding α -syn [14]. Since the discovery of SNCA, approximately 100 different genes or genetic loci have been found to be associated with PD to date [15]. Among these, high-penetrance monogenic forms of PD mainly include SNCA, VPS35, PRKN/PARK7/PINK1, while genes with variable penetrance primarily include LRRK2 and GBA [16]. Interactions between many common and rare genetic variants may also contribute to PD.

Patients with SNCA gene mutations are typically younger, experience rapid disease progression, exhibit swiftly advancing motor symptoms alongside prominent non-motor symptoms, and demonstrate a decline in cognitive ability within a short period [17]. VPS35 gene mutations can lead to mitochondrial dysfunction and autophagy defects, preventing the timely clearance of aggregated proteins and thereby promoting Lewy body formation, followed by damage to dopaminergic (DA) neurons [18]. PRKN, PINK1, and PARK7 are common autosomal recessive genes. Mutations in these genes result in impaired mitophagy and inflammatory responses, leading to the accumulation of abnormally aggregated proteins, exacerbating neuronal damage, and causing peripheral neuropathy and autonomic dysfunction. This manifests as gait and dystonia, tremor, and ataxia. However, PD caused by these genetic mutations usually progresses relatively slowly [19]. LRRK2 gene mutations are autosomal dominant. Mutated LRRK2

impairs vesicle trafficking, mitochondrial function, and autophagy, leading to significant α -syn accumulation and degeneration of nigral DA neurons. Patients present with symptoms such as tremor, dystonia, bradykinesia, cognitive decline, and psychiatric abnormalities [20]. GBA gene mutations cause dysfunction of the autophagy-lysosome pathway and reduced glucocerebrosidase (GCase) activity [21], resulting in α -syn aggregation and OS. This is followed by loss of nigrostriatal dopaminergic neurons, with patients exhibiting manifestations such as depression, sleep disturbances, and cognitive dysfunction [22].

Genetically caused PD is primarily divided into two forms: familial PD and sporadic PD. Familial PD, often referred to as Mendelian or monogenic PD, is generally rare but plays a crucial role in PD development due to its high penetrance. Genome-wide association studies (GWAS) have found that sporadic PD is believed to arise from interactions between the genome and the environment and is strongly associated with common genetic variants exhibiting low-penetrance effects [23]. Genetic variants account for approximately 25% of the risk factors for PD, and individuals with a family history of PD have a higher probability of developing the disease [24].

Environment and lifestyle

Mounting evidence suggests that environmental factors may contribute significantly to the prevalence of PD [25]. For instance, pesticides, organic solvents, agricultural work, well-water consumption, air pollution, traumatic head injury, and type 2 diabetes have been closely associated with the development of PD [11]. Furthermore, alterations in metal ion homeostasis within the brain also play a crucial role in PD pathogenesis [26]. Pesticides such as paraquat and rotenone can impair vesicular transport, neuronal function, lysosomal activity, and mitochondrial integrity. Long-term exposure to these agents may elevate the risk of PD [27].

MPTP, a well-established neurotoxin used in modeling PD, induces OS, aberrant α -syn aggregation, and dopaminergic neuronal death by inhibiting mitochondrial complex I [28]. This highlights the potential of environmental toxins to trigger PD through similar mechanisms. Research has shown that PM2.5 generated from biomass fuel activates the IL-17A/IL-17RA signaling pathway [29], leading to systemic and cerebral neuroinflammation, ultimately resulting in dopaminergic neuronal loss, α -syn pathology, and Parkinsonian motor and cognitive deficits [30]. Additionally, metabolic dysregulation of metal ions such as iron, copper, and manganese in the brain can catalyze Fenton reactions, generating reactive oxygen species that exacerbate OS and protein misfolding [31]. Interestingly, coffee and tea consumption have been found to potentially decelerate PD progression.

Caffeine in coffee may exert neuroprotective effects by blocking adenosine A2A receptors, thereby promoting dopaminergic neurotransmission. Tea, likely due to its antioxidant components, may also offer protective benefits to the organism [32].

Gut-Brain axis and microbiota in PD pathogenesis

In recent years, accumulating evidence suggests that the gut microbiota may play a significant role in the progression of PD. As the gastrointestinal tract is in direct contact with the environment, various environmental toxins can also impact the gut microbiome. PD patients often exhibit gut dysbiosis, characterized by an increase in *Akkermansia* and decreases in *Faecalibacterium* and *Roseburia* [33]. Such microbial imbalance may lead to impaired intestinal barrier function, inflammatory responses, and abnormal aggregation of α -syn within the enteric nervous system. This pathology can subsequently propagate to the central nervous system via the vagus nerve, potentially triggering PD [34].

The bidirectional communication pathway between the gut and the brain has long been a focus of research. A potential connection exists between gut bacteria and the brain, a bidirectional interaction referred to as the microbiota-gut-brain axis [35]. This axis comprises the central nervous system, autonomic nervous system, enteric nervous system, and the hypothalamic–pituitary–adrenal axis. Signals from the gastrointestinal tract can reach the central nervous system, while the central nervous system can relay signals back to the enteric nervous system [36]. This system serves as a critical interface for the interaction between microbiota and brain function [37].

Research by Challis et al. demonstrated that inoculation of pathogenic α -syn preformed fibrils into the gut of aged mice resulted in a marked increase of α -syn in the brainstem and the emergence of pathological changes in the central nervous system [38]. This indicates that α -syn can propagate from the gut to the brain. Conversely, injection of α -syn into the striatum also induced pathological manifestations within the enteric nervous system, demonstrating that α -syn propagation can occur not only from the gut to the brain but also from the brain back to the gut. These findings underscore the important role of the microbiota-gut-brain axis in the pathogenesis of PD.

Role of peripheral organs: spleen, liver, and skin

As a vital immune organ, the spleen has also been identified as a potential therapeutic target for neurodegenerative diseases. It influences central nervous system inflammation by regulating peripheral immune responses [39]. Research indicates that spleen dysregulation can activate immune cells such as M1 macrophages. Once activated, these macrophages promote the release

of pro-inflammatory factors like tumor necrosis factor- α (TNF- α), activate signaling pathways including NF- κ B, and drive neuroinflammation throughout the body and brain [40]. This process exacerbates dopaminergic neuron damage, accelerating the progression of PD [41].

Recent research has shown that the liver, a major detoxification organ, can also accumulate α -syn. Populations with hepatic α -syn accumulation exhibit a higher incidence of PD [42]. Thus, α -syn deposition in the liver may also contribute to PD pathogenesis. To date, multiple studies have observed deposits of α -syn and phosphorylated α -syn in skin biopsies from PD patients [43], with varying α -syn positivity rates across different biopsy sites [44]. Therefore, α -syn aggregation in the skin may play a significant role in PD development.

These peripheral organ–brain axis mechanisms collectively suggest that α -syn may originate in the periphery and propagate to the central nervous system through multiple pathways, thereby driving PD pathology. Intervention strategies targeting these axes—such as modulating gut microbiota, peripheral immunity, or hepatic metabolism—may offer novel directions for disease-modifying therapies in PD.

Pathogenesis of Parkinson's disease

The pathogenesis of PD arises from a highly integrated network rather than being driven by a single linear pathway. This network centers on the abnormal aggregation of α -syn as its core driver, which is deeply interconnected with major pathological modules including mitochondrial dysfunction, OS, neuroinflammation, and metabolic disturbances [45]. Specifically, α -syn pathology not only directly impairs mitochondrial function and activates neuroinflammation, but its own aggregation process is also markedly promoted by OS and inflammatory environments [46]. In turn, dysfunctional mitochondria generate excessive reactive oxygen species (ROS), which can directly induce neuronal damage and further drive the cascade of α -syn misfolding and neuroinflammatory amplification. Concurrently, pathological signals originating from peripheral systems such as the gut–brain axis and spleen–brain axis continuously propagate to the central nervous system, persistently exacerbating the pathological processes within the central network and ultimately leading to the progressive loss of nigral dopaminergic neurons [3]. A deep understanding of the pathogenic mechanisms of PD is of crucial importance for overcoming the limitations of current single-target therapies and for developing synergistic treatment strategies capable of simultaneously intervening at multiple key nodes (Fig. 1).

Abnormal aggregation of proteins

A-syn is a soluble, small-molecule protein consisting of 140 amino acids encoded by the SNCA gene, primarily localized at the presynaptic terminals of neurons. α -syn primarily consists of three domains: the N-terminus, the central hydrophobic NAC region, and the C-terminus. The N-terminus mainly maintains the helical structure of α -syn, and familial PD-associated mutations in the SNCA gene frequently occur in this region. The NAC region is the central domain of α -syn, where the conformational transition from α -helix to β -sheet structures primarily occurs, playing a crucial role in PD pathogenesis [47]. Under physiological conditions, α -syn exhibits no neurotoxicity. However, pathological states such as OS, genetic mutations, and iron overload can trigger abnormal accumulation of α -syn. Mutations and overexpression of the α -syn gene induce conformational changes, transforming soluble monomers into insoluble toxic aggregates. These aggregates further aggregate into more toxic fibrillar forms, ultimately forming LBs.

Notably, α -syn pathology is not an isolated process; it is both a consequence of other mechanisms and an active driver of disease progression. α -syn aggregates can spread within the central nervous system and exert cytotoxicity through multiple pathways, such as mitochondrial dysfunction, endoplasmic reticulum stress, and synaptic damage. This alters neuronal membrane potential and OS states, ultimately leading to neuronal death [23, 48, 49]. Additionally, PD pathogenesis primarily involves vesicular transport and the maintenance of lysosomes and mitochondria [50]. α -syn oligomers can damage mitochondrial complex I, impairing respiratory function, inducing selective oxidation of ATP synthase and mitochondrial lipid peroxidation, leading to mitochondrial dysfunction and subsequent cell death [51].

With aging, the functions of the UPS and autophagy-lysosomal pathway gradually decline, leading to a significant increase in α -syn expression in the substantia nigra region and resulting in metabolic disorders [52]. In PD patients, massive accumulation of α -syn in the brain forms LBs. Misfolded α -syn exhibits properties similar to prions. As the disease progresses, LBs proliferate within the brain and migrate to other regions, causing multi-regional brain degeneration while simultaneously activating the NLRP3 inflammasome to trigger neuroinflammation [53].

Oxidative stress

Oxidative stress plays a critical role in the progression of PD. In current research, one of the most important pathogenic mechanisms of PD is the DA neurons caused by OS. During aging, OS represents a primary process that

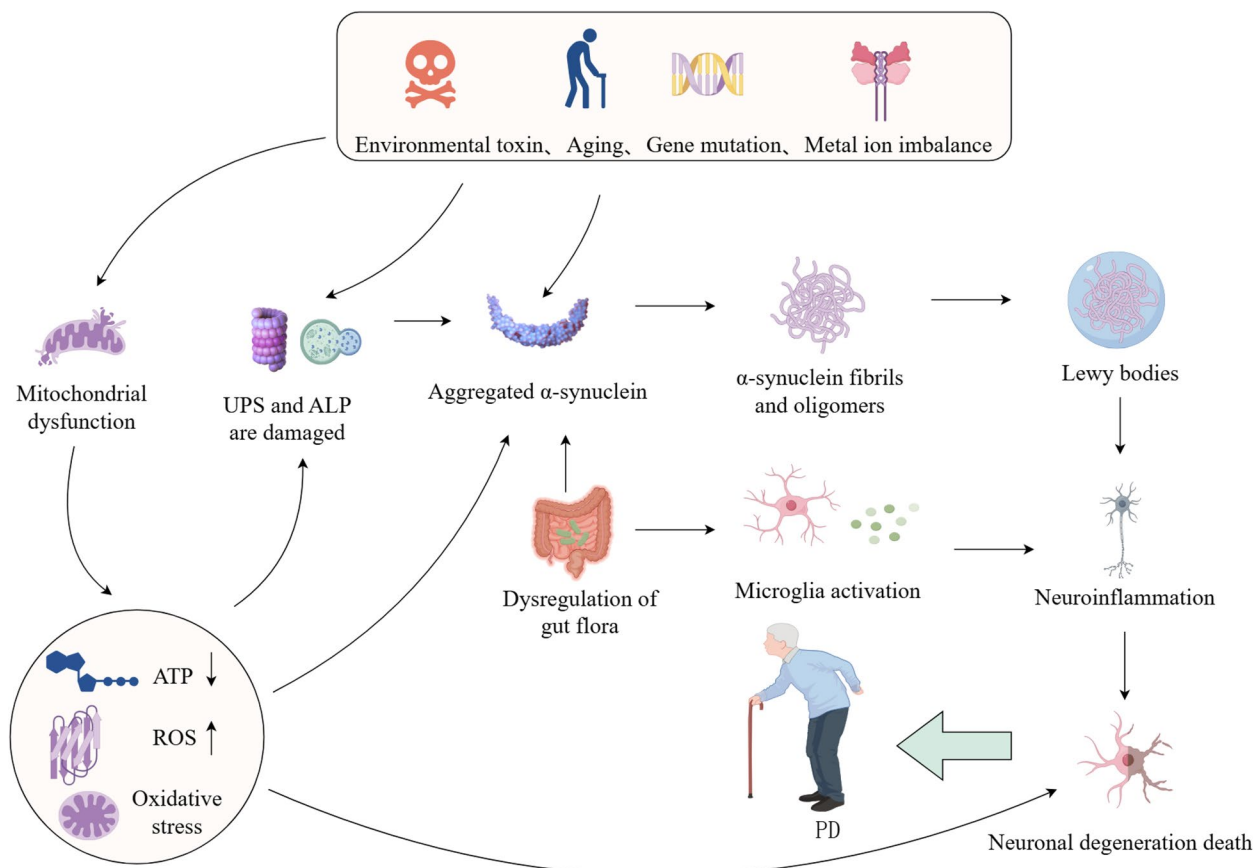


Fig. 1 Molecular mechanisms contributing to Parkinson's disease. The figure illustrates the interplay between the diverse pathogenic mechanisms of PD and their contribution to the disease's progression. A number of factors, including environmental toxins, the process of aging, genetic mutations, imbalances in metal ions, and others, can contribute to the development of mitochondrial dysfunction, damage to the ubiquitin–proteasome system (UPS) and autophagy-lysosome system, and the aggregation of α -syn; Mitochondrial dysfunction results in a reduction in adenosine triphosphate (ATP) synthesis, an increase in ROS production and OS. This, in turn, leads to damage to the UPS and autophagy-lysosomal system, as well as the aggregation of α -syn, denaturation of neurons and their subsequent death. Intestinal dysbacteriosis has been demonstrated to cause α -syn aggregation and microglia activation, which in turn induces neuroinflammation. The formation of α -syn aggregates into fibrils and LBs has been linked to the onset of neuroinflammation, which in turn leads to neuronal deformation and death. This process plays a pivotal role in the development of PD

directly impairs the central nervous system [54]. ROS are highly reactive molecules mainly involved in various redox reactions in the body and serve important defensive functions. However, excessive generation of ROS can lead to OS [55].

Under normal physiological conditions, excess free radicals produced by brain metabolism can be cleared by endogenous antioxidants such as superoxide dismutase (SOD) and glutathione (GSH), maintaining a balance between ROS production and elimination [56]. In PD patients, the levels of antioxidants in the brain are reduced, and both the activity of antioxidant enzymes and the capacity to scavenge free radicals are diminished. This disrupts the body's redox balance, triggering OS. Consequently, free radicals accumulate substantially in

the substantia nigra, and cytotoxic substances that cannot be cleared aggregate within neurons, ultimately leading to cell death and DA neurons [57].

Additionally, when aged cells excessively accumulate in the body, the release of highly reactive molecules such as reactive oxygen species and reactive nitrogen species (RNS) increases, inducing OS. This impairs cellular functions and further exacerbates mitochondrial dysfunction [58]. Importantly, OS engages in profound bidirectional interactions with other pathogenic mechanisms. On one hand, ROS can directly modify α -syn, accelerating its abnormal aggregation; on the other hand, it is both a major product of mitochondrial dysfunction and a contributor to further mitochondrial damage [59].

Mitochondrial dysfunction

Mitochondria serve as the primary intracellular source of ROS during normal aging. The ATP generated by mitochondria powers neural activity through oxidative phosphorylation in the mitochondrial electron transport chain and helps maintain intracellular homeostasis. Mitochondrial dysfunction arises from the combined effects of environmental and genetic factors and is closely linked to the onset of PD [60]. Such dysfunction leads to altered mitochondrial dynamics, impairment of the mitochondrial electron transport chain, increased ROS production, and consequently exacerbates OS [61]. This disrupts cellular energy metabolism, reduces ATP synthesis, and results in progressive cellular dysfunction and neurodegeneration. ROS derived from mitochondria can also impair lysosomal and proteasomal function and promote abnormal aggregation of α -syn [62].

Moreover, the balance of metal ions in the body plays a vital role in sustaining normal neurophysiological activity. With advancing age, certain changes in metal ion levels occur, and alterations in the content of metal ions in the brains of PD patients contribute to disease pathogenesis. Studies indicate that elevated levels of metals such as Fe, Cu, Mn, and Zn can damage the UPS and are associated with OS, mitochondrial dysfunction, and protein misfolding [63]. Furthermore, redox-active metals like copper and iron can catalyze the Fenton reaction, generating substantial amounts of ROS, which in turn induces OS and ultimately leads to cell death. Increased ROS also promotes iron oxidation, producing cytotoxic hydroxyl radicals from hydrogen peroxide and aggravating α -syn aggregation. Research shows that elevated manganese levels enhance intracellular ROS formation, inhibit mitochondrial complexes I and II of the electron transport chain, disrupt mitochondrial function, and cause dopaminergic cell death [64].

Additionally, Zn^{2+} accumulation promotes the death of dopaminergic neurons, and Zn^{2+} deposition has been observed in degenerating dopaminergic neurons in PD mouse models [65]. Thus, imbalance in metal ions may facilitate the progression of PD. Meanwhile, mitochondrial dysfunction impairs ATP-dependent lysosomal and proteasomal functions, weakens the cellular self-clearing capacity, and thereby indirectly contributes to α -syn accumulation. Consequently, mitochondria are not only a source of energy but also a central hub connecting various pathogenic pathways [66].

Neuroinflammation

Neuroinflammation is a critical factor in the pathogenesis of PD. Although neuroinflammation serves as a neuroprotective mechanism, prolonged neuroinflammation can lead to neurotoxicity and subsequent

neurodegeneration. Chronic inflammation directly or indirectly promotes the progression of PD. Under normal physiological conditions, microglia and astrocytes maintain homeostasis in the central nervous system by releasing neurotrophic factors and synaptic glutamate [67]. Under pathological stimulation, overactivation of microglia leads to the release of large amounts of pro-inflammatory factors, ROS, and nitric oxide (NO), triggering uncontrolled inflammatory responses that ultimately result in neuronal damage and sustained neuroinflammation [68].

Numerous studies have shown that, particularly in microglia, the activation of inflammasomes induces chronic neuroinflammation, which in turn initiates a cascade of reactions that damage dopaminergic neurons, leading to PD-like symptoms. Concurrently, activation of the NLRP3 inflammasome has been observed in Parkinson's patients, suggesting that inflammasomes play a significant role in neuroinflammation [69, 70]. In the brains of PD patients, levels of microglia and astrocytes are abnormally elevated. Activated glial cells release various chemokines and inflammatory mediators, including interleukin-1 β (IL-1 β), IL-4, IL-6, TNF- α , and interferon- β , which disrupt the blood-brain barrier. These pro-inflammatory factors stimulate astrocytes, contributing to the degeneration and death of dopaminergic neurons and oligodendrocytes [71]. Inflammatory cytokines not only directly damage neurons but also further impair mitochondrial function, inhibit proteasome activity, and create a favorable environment for α -syn aggregation, thereby exacerbating proteostasis imbalance and OS [72].

Disturbance of metabolism

Recent studies have indicated that glucose metabolism, lipid metabolism, and amino acid metabolism are closely associated with the development of PD. With aging, reduced cellular metabolic activity leads to the accumulation of ROS, resulting in abnormal protein aggregation and organelle damage, which ultimately cause UPS dysfunction and impaired autophagy. These processes promote cell death, neuronal loss, and subsequently trigger PD [49, 73].

Abnormal glucose metabolism is closely associated with PD. Compared to healthy individuals, PD patients exhibit a higher prevalence of diabetes. Furthermore, an increased risk of PD onset has been observed in patients with hyperglycemia and diabetes [74]. Hyperglycemia can impair PD-related mesolimbic cortical pathways and nigrostriatal motor pathways, with the nigrostriatal motor pathway undergoing preferential degeneration [75]. Research indicates that hyperglycemia exacerbates PD pathology through multiple mechanisms. On the one hand, it induces OS by upregulating the transcription

factor Nuclear Factor E2-related Factor 2 (Nrf2) and downregulating its inhibitor Keap1 [76], while also triggering neuroinflammation that causes OS and mitochondrial dysfunction in dopaminergic neurons [77]. On the other hand, it promotes α -syn aggregation by increasing thioredoxin-interacting protein (TXNIP), thereby inhibiting autophagy flux and mitochondrial autophagy, which accelerates dopaminergic neuron apoptosis and worsens PD progression [78].

Lipids play crucial roles in neuronal structure and physiological functions and are essential for the development and maintenance of the central nervous system [79]. Growing evidence indicates that lipid metabolism disorders are closely associated with neurodegenerative diseases [80]. Abnormal intracellular lipid accumulation induces mitochondrial dysfunction, reduces mitochondrial quantity, and further promotes lipid accumulation [81]. Studies show that PD risk genes such as GBA, LRRK2, and PARK2 are also linked to lipid metabolism [82]. Lipid accumulation impairs mitochondrial function, while mitochondrial dysfunction, in turn, exacerbates lipid metabolism disturbances [73]. Concurrently, lipid metabolism disorders can lead to impaired neuronal function, lysosomal dysfunction, and autophagic deficits, promoting neuronal death and contributing to various neurological diseases [83].

Research indicates that abnormal amino acid metabolism can disrupt synaptic signaling, leading to excitotoxicity and neuronal death. Additionally, oxidative toxicity induced by amino acid metabolism abnormalities is closely associated with ferroptosis [84]. In PD models, the use of dependent sodium channel blockers reduces glutamatergic hyperactivity and inhibits abnormal glutamate release, suggesting that amino acid metabolism disturbances may contribute to neurotoxicity and oxidative toxicity, ultimately resulting in dopaminergic neuron death [85]. Cysteine, a constituent amino acid of glutathione, is critical for antioxidant defense. Reduced cysteine uptake leads to decreased glutathione (GSH) levels, and low GSH concentrations induce OS, mitochondrial dysfunction, protein oxidative damage, and neurodegeneration, thereby promoting PD [86].

These metabolic pathways interact deeply with core mechanisms such as OS, mitochondrial dysfunction, and abnormal α -syn aggregation, collectively forming the complex metabolic background of PD. This integrated perspective provides new insights for understanding disease pathogenesis and developing novel therapeutic strategies.

Other

Recent studies have revealed that multiple peripheral-central pathways, including the gut-brain axis, spleen-brain axis, liver-brain axis, and skin-brain axis, collectively

constitute a complex pathological network in PD. These axes interact with the central nervous system through distinct molecular mechanisms and play significant roles in the initiation, propagation, and progression of PD.

The role of the gut microbiota-brain axis in PD pathogenesis is increasingly well-defined. PD patients exhibit characteristic gut microbiota dysbiosis. Research indicates that as PD severity increases, *Akkermansia* levels rise, whereas *Faecalibacterium* and *Roseburia* decrease with disease progression [87]. Notably, *Faecalibacterium* and *Roseburia* are short-chain fatty acid (SCFA)-producing genera that confer beneficial protective effects on the gut, and their reduction may impact the cognitive function and mood of PD patients [88]. The accumulation of *Akkermansia* promotes intestinal mucosal barrier damage and gut inflammation, facilitating abnormal α -syn aggregation within the intestine, which can ultimately lead to systemic inflammation [89]. SCFAs are crucial for modulating intestinal epithelial barrier and blood-brain barrier integrity, inflammatory responses, and endocrine signaling, playing a vital role in communication along the microbiota-gut-brain axis. While fecal SCFA levels are decreased in PD patients, their levels in plasma, urine, and saliva are elevated. Reduced intestinal SCFA levels are closely associated with worse cognitive and motor function in PD patients [90].

Furthermore, prior to the onset of motor symptoms, over half of PD patients experience non-motor symptoms such as chronic constipation due to enteric nervous system dysfunction [91]. Constipation is the most common gastrointestinal dysfunction in PD. Increased *Akkermansia* is closely linked to constipation, while constipation severity correlates significantly with reduced *Blautia* and *Faecalibacterium*. Gut microbiota dysbiosis in PD patients may increase intestinal permeability, exposing the enteric plexus to various toxins. The overall gut microbial profile in PD is characterized by decreased SCFAs and elevated lipopolysaccharide (LPS) levels [92]. This leads to impaired microglial and astrocytic function, exacerbating neuroinflammation and increasing blood-brain barrier (BBB) permeability. Additionally, microglial activation promotes abnormal α -syn fibril aggregation, augments OS, and results in neurodegeneration and gastrointestinal dysfunction, thereby accelerating PD progression. Moreover, the gut microbiota can produce microbial amyloid proteins, which not only aggravate inflammatory responses but also accelerate α -syn aggregation in the gut. These pathological proteins can then propagate to the central nervous system via the vagus nerve, exacerbating PD development [93]. The therapeutic implications of gut microbiota dysbiosis in PD extend beyond symptom relief and may involve modulating gut-brain axis mechanisms to delay disease progression.

Exploring the deep-seated connections between dysbiosis and PD could thus open new avenues for PD treatment.

Additionally, the spleen is a vital peripheral immune organ in the body, harboring diverse immune cells including macrophages, lymphocytes, and dendritic cells, which collectively form a complex immune micro-environment. Reports indicate that in MPTP-induced PD mouse models, a significant increase in splenic macrophages is observed. The rise in M1 macrophages activates the NF- κ B signaling pathway, leading to elevated levels of proinflammatory factors [94]. This suggests that the spleen, particularly its macrophage population, may participate in neuroimmune regulation of Parkinson's disease by bridging peripheral and central inflammation.

Research on the liver-brain axis has uncovered the potential significance of the liver in PD. Evidence indicates that α -syn deposition is observed in the livers of both PD models and patients, showing an age-related distribution pattern [42, 95]. Notably, brain-derived α -syn has been found to propagate to the liver, suggesting bidirectional communication of pathological proteins between organs [42]. These findings indicate that the livers of PD mouse models are affected by α -syn aggregation, and α -syn may propagate from the brain to the liver, subsequently inducing hepatic inflammation [42]. This suggests that the liver may be involved in the pathological process of PD, influencing its progression.

Compared with healthy adults, PD patients exhibit significant α -syn deposition in cutaneous autonomic nerve fiber terminals, with notably higher levels in the scalp and cervical skin than in other sites [96]. Growing evidence suggests that α -syn accumulation in cholinergic and adrenergic skin fibers holds important diagnostic value for idiopathic PD [97]. The pathology of α -syn aggregation and dopaminergic neuron loss in the peripheral skin resembles that in the central nervous system, indicating a strong correlation between cutaneous and CNS pathology [98].

These peripheral-central axes provide multiple points of origin for α -syn propagation through mechanisms such as immune activation, pathological protein deposition, and inflammatory signaling, thereby systemically exacerbating central nervous system damage. This systemic perspective opens new pathways for PD prevention and treatment, suggesting that targeting gut microbiota, modulating splenic immunity, clearing hepatic α -syn, or utilizing skin biopsies for early diagnosis may become important future therapeutic strategies.

Parkinson's disease medications

Although many novel therapeutic technologies for PD are under development, current treatment remains primarily focused on symptomatic management aimed at

correcting the imbalance of neurotransmitters in the brain. The primary pharmacological target of anti-PD drugs is the dopaminergic system (Fig. 2), with dopamine-mimetic agents and anticholinergic drugs being the most commonly used in clinical practice (Table 1). Currently, L-DOPA remains the most effective medication for PD. However, long-term administration of L-DOPA is associated with the emergence of adverse effects such as motor complications, which has prompted researchers to actively explore new therapeutic approaches and develop novel anti-PD agents to address disease progression.

Existing drugs for clinical use

L—DOPA

The primary cause of PD is the deficiency of dopamine (DA) in the nigrostriatal pathway. In PD patients, the imbalance between dopamine and acetylcholine (ACh) levels in the striatum leads to a series of clinical symptoms. Consequently, traditional therapeutic strategies focus on replenishing cerebral dopamine. L-DOPA, as a dopamine precursor, remains the most effective medication for PD [99]. Combined with peripheral dopa decarboxylase inhibitors in fixed-dose formulations, it is converted into dopamine after crossing the blood–brain barrier, effectively compensating for the dopaminergic deficit in the nigrostriatal pathway and significantly improving core motor symptoms such as bradykinesia, rigidity, and tremor [100], long-term use is associated with motor complications including wearing-off phenomena, dyskinesias, and "on–off" fluctuations, as well as peripheral dopaminergic adverse effects such as nausea, vomiting, and hypotension [101, 102]. Despite its well-established efficacy, L-DOPA therapy faces significant limitations. Compared with other medications, while L-DOPA has a rapid onset of action, its long-term use carries a higher risk of complications.

Recent clinical trials aiming to slow disease progression have failed to meet their primary endpoints, suggesting that dopamine replacement alone is insufficient to halt disease advancement. Therefore, key challenges for future research include balancing short-term symptom control with long-term complication risks, developing non-dopaminergic therapies to address non-motor symptoms, and exploring combination drug strategies to achieve synergistic therapeutic effects.

Dopamine agonists

Dopamine receptor agonists are one of the commonly used medications for the treatment of PD. Their mechanism of action involves direct stimulation of postsynaptic dopamine receptors in the striatum, contributing to both neuroprotective and reparative effects. These agents demonstrate considerable efficacy in managing PD

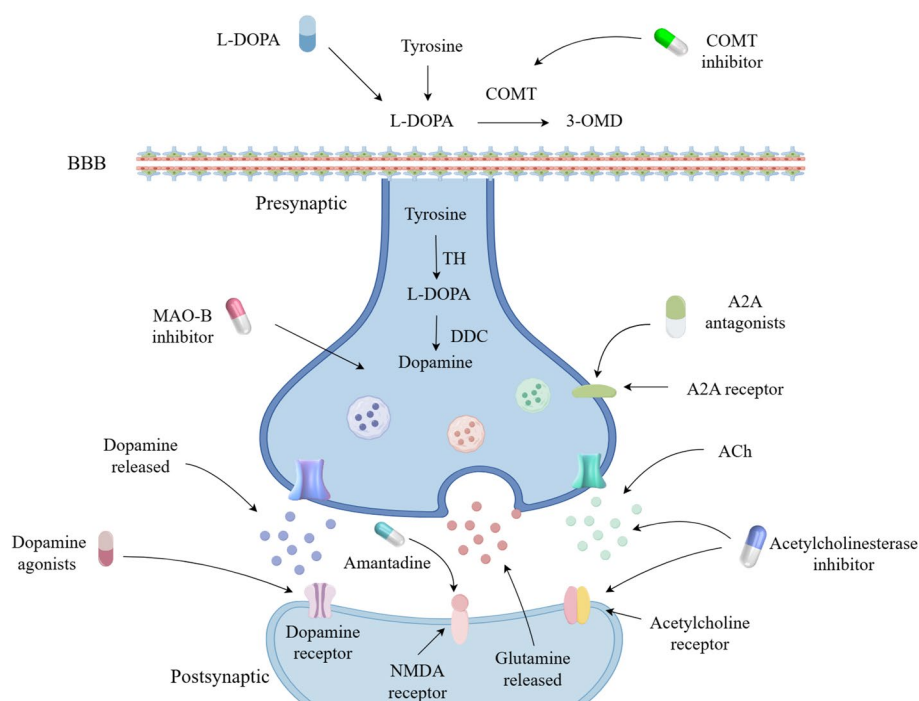


Fig. 2 The action sites of different anti-PD drugs. This diagram describes the mechanism of action and the release of neurotransmitters of various anti-PD drugs. BBB=blood–brain barrier; TH=Tyrosine hydroxylase; 3-OMD=3-O-Methyldopa; DDC=DOPA decarboxylase; MAO-B= Monoamine oxidase B; COMT= Catechol-Omethyl transferase; NMDA= N-Methyl-D-aspartic acid

Table 1 Current clinical drugs for the treatment of PD

Drug category	Drugs	target	efficacy
Dopamine precursor drugs	L-DOPA	DR	Improve motor and non-motor symptoms
DAs	Bromocriptine, cabergoline, lisuride, pergolide, priribedil、 pramipexole, ropinirole, rotigotine, apomorphine	DR	Improve movement disorders
MAO-B inhibitor	Selegiline, rasagiline, safinamide	MAO-B	Improve motor symptoms
COMT inhibitor	Tocapone, Entacapone, Opicapapone	COMT	Improve the motion fluctuation caused by L-DOPA
Amantadine	Amantadine	NMDAR	Relieve movement disorders, Reduce the off period time
Anticholinergics	Trihexyphenidyl、 Benzat ropine	MACHR	Treatment of PD patients with tremor
Adenosine A2A receptor antagonist	istradefylline	Adenosine A2A recep-tor	Reduce the off period time, improve movement disorders

DR Dopamine Receptor, Das Dopamine agonists, MAO-B Monoamine oxidase B, COMT Catechol-O-methyltransferase, NMDAR N-methyl-D-aspartate receptor; MACHR= Muscarinic Acetylcholine Receptor

symptoms. In early-stage PD, they can be used as monotherapy to delay the introduction of L-DOPA, while in advanced stages, their combination with L-DOPA helps mitigate associated motor complications [103].
Currently, clinical use of dopamine agonists has shifted from earlier ergot derivatives to safer non-ergot agents. These include oral formulations such as pramipexole and ropinirole, the transdermal patch rotigotine,

and subcutaneous apomorphine for acute "off"-period rescue [104]. In addition to effectively improving motor symptoms and reducing "off" time, some agents—notably pramipexole—have shown benefits for non-motor symptoms such as depression and sleep disturbances [105, 106]. Orthostatic hypotension, dizziness, drowsiness, and hallucinations are common adverse reactions [107, 108].

Although dopamine agonists offer advantages in delaying motor complications, their overall efficacy remains inferior to that of L-DOPA, and they are associated with a higher incidence of behavioral adverse events. These limitations drive ongoing research efforts aimed at developing next-generation agents with improved receptor selectivity profiles to enhance their overall clinical utility.

MAO-B inhibitors

Monoamine oxidase (MAO) is an enzyme embedded in the outer mitochondrial membrane that catalyzes the oxidative deamination of monoamine neurotransmitters and tyramine. It exists in two subtypes, MAO-A and MAO-B, with MAO-B being clinically utilized in the treatment of PD [109]. MAO-B is a key enzyme involved in the catabolism of dopamine. MAO-B inhibitors primarily work by inhibiting the metabolism of dopamine in the brain, reducing its deamination, thereby increasing both the amount and duration of dopamine action. Additionally, they may prevent the formation of neurotoxins, thereby slowing the progression of neuronal degeneration [110].

In clinical practice, the value of these agents is reflected in two main aspects: as monotherapy in early disease they can delay the need for L-DOPA, and as adjunctive therapy in moderate to advanced stages they effectively reduce "off" time associated with L-DOPA. The development of this drug class has evolved through three generations. The first-generation agent selegiline, while effective in improving symptoms, has relatively low selectivity, often causes side effects such as insomnia, and requires attention to drug interactions [111, 112]. The second-generation agent rasagiline exhibits higher selectivity and bioavailability, has a milder side-effect profile, does not require dietary restrictions, and may offer potential benefits in slowing disease progression alongside improving motor symptoms [113]. The third-generation agent safinamide achieves high selectivity and reversible inhibition, featuring a unique dual mechanism of action [114]. It not only inhibits MAO-B but also modulates glutamate release, significantly reducing the incidence of motor fluctuations [115, 116].

Although MAO-B inhibitors are generally well tolerated, their clinical application still faces key challenges. Future research should focus on developing new compounds with enhanced selectivity and clearly demonstrated disease-modifying effects, as well as exploring their potential for intervention in the prodromal phase of PD.

COMT inhibitors

COMT inhibitors primarily function by inhibiting COMT, thereby increasing the concentration of L-DOPA in the brain and preventing its peripheral degradation

into 3-O-methyldopa (3-OMD). This effectively prolongs the plasma half-life and duration of action of L-DOPA, which plays a crucial role in the treatment of PD [117]. By protecting L-DOPA from metabolism by peripheral COMT enzymes, COMT inhibitors are important therapeutics for managing motor fluctuations induced by long-term L-DOPA therapy. They improve motor fluctuations through extending central dopamine activity [118]. The core clinical value of this drug class lies in their use as adjuncts to L-DOPA, significantly prolonging patients' "on" time, shortening "off" time, and potentially reducing the total daily dose of L-DOPA.

Currently, the main agents used clinically include tolcapone, entacapone, and opicapone. Among them, tolcapone, the first COMT inhibitor, demonstrates notable efficacy due to its combined peripheral and central action [119]. However, its potential hepatotoxicity risk necessitates strict liver function monitoring, limiting its use as a first-line agent [120]. Entacapone, the most commonly used peripherally selective inhibitor, carries no hepatotoxicity risk but must be administered concomitantly with each L-DOPA dose [121]. Its common side effects include dyskinesia and urine discoloration [122]. The newest generation agent, opicapone, is also a peripherally selective inhibitor. It achieves once-daily dosing owing to its prolonged enzyme inhibitory capacity [123, 124], greatly improving medication convenience, though it is associated with a relatively higher incidence of dyskinesia [125].

Overall, the application of COMT inhibitors requires balancing the clinical benefit of extended "on" time against the risk of exacerbating dyskinesia. Future research should focus on optimizing dosing strategies, clarifying the mechanisms underlying individual differences in treatment response, and exploring their potential in combination with novel therapies, aiming to provide better management strategies for motor fluctuations.

Amantadine

Amantadine is a noncompetitive antagonist of the glutamate receptor N-methyl-D-aspartate receptor (NMDAR), exerting both direct and indirect effects on glutamatergic and dopaminergic signaling. By inhibiting NMDAR, it can control dyskinesia and is used in the treatment of L-DOPA-induced dyskinesia [126]. Amantadine possesses not only antiglutamatergic but also anticholinergic properties. It promotes the release of dopamine by stimulating the terminals of nigrostriatal dopaminergic neurons and reduces neuronal reuptake of dopamine, thereby increasing dopamine levels and exerting anti-parkinsonian effects [127].

Its therapeutic value is primarily demonstrated in two aspects: significantly improving dyskinetic symptoms,

and effectively prolonging "on" time while shortening "off" time. This makes it the only drug currently approved by the FDA for treating both types of motor complications [128]. Regarding safety, amantadine is generally well-tolerated, with common adverse reactions including hallucinations, dizziness, and constipation, most of which are mild to moderate [129, 130].

However, important limitations remain in its clinical application. Its exact mechanism of action has not been fully elucidated, and its efficacy is primarily evident at the level of symptom control, lacking evidence for disease-modifying effects. Future research needs to further explore its neuromodulatory mechanisms and develop synergistic application strategies with other treatments to better realize its unique value in PD management.

Anticholinergics

Anticholinergic drugs were the first class of medications used clinically for the treatment of PD. Trihexyphenidyl and Benztropine are commonly used anticholinergic agents in clinical practice. Their mechanism of action involves blocking cholinergic receptors, thereby reducing the effect of acetylcholine in the nigrostriatal pathway to restore the balance between dopamine and cholinergic neurotransmitters, which underlies their therapeutic effect [131]. Clinically, anticholinergic drugs are often used in combination with L-DOPA or dopamine receptor agonists, primarily for the treatment of PD patients with tremor [132].

The antagonism of acetylcholine can lead to adverse reactions such as cognitive impairment, confusion, blurred vision, hallucinations, urinary retention, and constipation. Cognitive impairment can severely affect patients' quality of life and is also a significant factor contributing to increased mortality in PD patients [133]. Therefore, anticholinergic drugs are generally recommended only for younger patients without cognitive impairment who have poorly controlled tremor [134].

Due to the numerous safety concerns associated with anticholinergic drugs, their clinical application is significantly limited. Future research should focus on elucidating the precise mechanisms underlying PD tremor and developing novel therapeutic strategies with regional specificity that can effectively control tremor while avoiding systemic cholinergic side effects.

Adenosine A2A receptor antagonist

In recent years, adenosine A2A receptor antagonists, as non-dopaminergic compounds capable of modulating striatopallidal output to improve motor disability, have been extensively studied [135]. In 2019, the US FDA approved the adenosine A2A receptor antagonist istradefylline as an adjunctive treatment to L-DOPA/carbidopa

in adult PD patients experiencing OFF episodes or the wearing-off phenomenon.

Its mechanism of action involves blocking A2A receptors within the indirect pathway of the basal ganglia, thereby modulating the function of motor circuits [136]. This agent is well-tolerated and easy to use, effectively reducing patients' OFF time without significantly exacerbating dyskinesia. Common adverse reactions observed clinically include dyskinesia, somnolence, orthostatic hypotension, nausea, and hallucinations; however, its overall tolerability profile remains favorable [137, 138]. Future research should explore combination strategies of this drug with other therapies to maximize its therapeutic value.

Emerging treatments

Inhaled Levodopa (CVT-301)

To address the issue of OFF episodes in PD patients despite oral L-DOPA administration, an inhaled L-DOPA formulation, CVT-301, has been developed. Utilizing dry powder technology delivered via a breath-actuated inhaler, it bypasses gastrointestinal degradation of L-DOPA. The drug is rapidly absorbed through the alveolar-capillary network and enters the central nervous system, leading to a quicker onset of action [139]. Compared to oral administration, it avoids impaired drug absorption associated with the gastrointestinal tract, thereby rapidly elevating plasma L-DOPA concentrations [140].

Studies have shown that inhalation of CVT-301 significantly increases ON time compared to placebo. Furthermore, it not only effectively reduces the frequency of OFF episodes and improves motor symptoms but also demonstrates a favorable safety and tolerability profile, with no serious adverse events reported. Common adverse reactions include nausea, cough, sputum discoloration, throat irritation, and other respiratory symptoms [141]. As the inhaled L-DOPA is delivered intranasally to the lungs and enters the systemic circulation, it carries a risk of bronchospasm and is not suitable for patients with chronic lung diseases [142].

In terms of clinical application, its mechanism of action defines it as an important tool for symptom management rather than a disease-modifying therapy. As an on-demand treatment, its duration of effect is limited, and it cannot replace long-term strategies such as L-DOPA-carbidopa intestinal gel or continuous subcutaneous apomorphine infusion, which aim to provide stable plasma drug concentrations. Overall, this formulation offers a unique and clinically implemented new option for the symptomatic management of PD, filling a therapeutic gap for rapid rescue of OFF episodes. However, its optimal usage strategy and long-term benefits still require further clarification in broader real-world settings.

Levodopa–carbidopa intestinal gel

To improve the utilization of L-DOPA, address its short half-life, and alleviate patients' motor symptoms, several different L-DOPA formulations have been developed. L-DOPA-Carbidopa Intestinal Gel (LCIG) is primarily administered via a portable infusion pump connected to a percutaneous endoscopic gastrojejunostomy (PEG-J) tube. This system delivers the drug directly into the jejunum, bypassing the stomach to prevent interference with absorption, thereby enhancing drug bioavailability and increasing dopamine levels in the body [143].

Studies have shown that after treatment with LCIG in advanced PD patients, both motor and non-motor symptoms were significantly improved, with concurrent enhancements in mood and sleep quality [144]. Multiple other studies have confirmed the efficacy of long-term LCIG use in PD patients. LCIG not only improves dyskinesia and sleep issues but also reduces OFF time and increases ON time [145]. It alleviates motor fluctuations without increasing troublesome dyskinesia. Significant improvements have been observed in PD patients' Unified Dyskinesia Rating Scale and Non-Motor Symptoms Scale scores, thereby enhancing patients' quality of life and reducing caregiver burden [146, 147].

However, as a highly invasive therapy, its clinical application necessitates a comprehensive risk–benefit evaluation. Adverse events are primarily associated with the PEG-J tube, with the most common complications being tube dislocation and stoma site infection [148]. Consequently, although LCIG represents a crucial therapeutic option for advanced PD, its widespread clinical use heavily depends on careful patient selection, management by specialized teams, and a holistic consideration of efficacy, safety, and cost-effectiveness.

Subcutaneous apomorphine infusion

Apomorphine is a dopamine agonist with affinity for various dopamine receptor subtypes, characterized by rapid absorption and a short half-life. To address the limitations of oral medications, continuous subcutaneous apomorphine infusion (CSAI) has been developed as a treatment for PD, primarily administered via an assistive device [149]. Studies have demonstrated that PD patients treated with CSAI exhibit a significant reduction in motor fluctuations, and favorable effects are also observed in patients with impulse control disorders [150]. Compared to placebo, CSAI markedly reduces daily OFF time, improves both motor and non-motor symptoms, and demonstrates a favorable safety and tolerability profile. Further research indicates that CSAI is particularly effective for patients with motor fluctuations and severe insomnia [151]. Treatment with CSAI effectively alleviates dyskinesia in PD patients while also

significantly improving their non-motor symptoms [152]. Additionally, it can reduce the daily dosage of L-DOPA, thereby minimizing motor complications. As apomorphine is administered subcutaneously, common adverse reactions include local symptoms such as subcutaneous nodules and erythema.

Potential therapies

With the deepening understanding of the pathological mechanisms underlying PD, current therapeutic strategies have shifted from purely dopaminergic replacement toward disease-modifying interventions targeting core pathological processes. This section focuses on disease-modifying strategies based on non-dopaminergic mechanisms, encompassing multiple directions such as regulation of protein homeostasis, restoration of brain iron metabolism balance, suppression of neuroinflammation, and improvement of energy metabolism. These potential therapies exert specific effects by targeting abnormal α -syn aggregation, correcting brain iron dyshomeostasis, modulating cellular signaling pathways, among other mechanisms, and have demonstrated significant neuroprotective effects in preclinical studies (Table 2). Some candidates have progressed into clinical trials and show promising therapeutic potential, marking a transition toward multitarget disease-modifying approaches in PD treatment and offering new hope for patients.

α -Synuclein misfolding inhibitors

The misfolding of α -syn plays a critical role in the pathogenesis of PD. Accordingly, therapeutic strategies targeting this mechanism have led to the development of α -syn misfolding inhibitors such as NPT200-11. Minzasolmin (UCB0599), a single enantiomer of NPT200-11, is an orally bioavailable small molecule that inhibits the early stages of α -syn misfolding, thereby preventing pathological α -syn aggregation [153].

Studies have demonstrated that minzasolmin exhibits substantial blood–brain barrier penetration, rapidly distributing throughout the brain following oral administration [154]. Research in PD animal models has shown that NPT200-11 reduces abnormal α -syn accumulation, suppresses astrocytic activation—thus attenuating neuroinflammation—and normalizes striatal dopamine transporter (DAT) levels, alleviating motor deficits in PD mice [155]. Phase I clinical trials of UCB0599 further confirmed its ability to cross the blood–brain barrier rapidly and distribute into cerebrospinal fluid, with predominant localization in white and gray matter. Notably, food intake did not significantly affect its pharmacokinetics, and the compound demonstrated a favorable safety and tolerability profile [156]. Currently, Phase II clinical studies of UCB0599 are underway.

Table 2 Research progress of potential therapeutic drugs for PD

Drugs	Trial number	Development stage	Effect
α -syn misfolding inhibitor			Inhibition of α -syn misfolding
Minzasolmin (UCB0599)	NCT04658186	Phase II(2020)	
c-Abl inhibitor			inhibition of c-Abl, inhibition of neurodegeneration
Nilotinib	NCT02954978	Phase II (2017)	
Vodobatinib (K0706)	NCT03655236	Phase II (2019)	
Risvodeltinib (IKT-148009)	NCT05424276	Phase II (2023)	
GLP-1 receptor agonists			Clear pathological α -syn, reduce neuroinflammation and movement disorders
Exenatide	NCT04232969	Phase III (2020)	
Liraglutide	NCT02953665	Phase II (2017)	
Semaglutide	NCT03659682	Phase II (2019)	
Lixisenatide	NCT03439943	Phase II (2018)	
Statin			Reduce the accumulation of α -syn and reduce neuroinflammation
Lovastatin	NCT03242499	Phase II (2017)	
Simvastatin	NCT02787590	Phase II (2016)	
Iron chelators			Reduce iron deposition in the substantia nigra, improve motor symptoms
Deferiprone	NCT02655315	Phase II (2016)	
Others			
Ambroxol	NCT02914366	Phase II(2016)	Increase the activity of GCase, decrease the level of α -syn
Terazosin	—	—	Alleviate motor symptoms, reduce the risk of PD dementia
Ginsenoside Rg1	—	—	Reduce neuroinflammation
Melatonin	—	—	Improve sleep disorders and motor defects

Research progress of potential therapeutic drugs for PD. This table summarizes ongoing and recently completed clinical trials of potential disease-modifying therapies for PD. Data were collected from ClinicalTrials.gov. Only interventional studies (Phase I–III) with available identifiers were included. Search Keywords: PD, Parkinsonism, Lewy body dementia, drug therapy, medication

Its disease-modifying potential lies in the early intervention of the α -syn aggregation cascade, distinguishing it from conventional symptomatic treatments. However, successful clinical development still faces key challenges, including the need to determine optimal dosing and treatment duration, identify validated biomarkers to accurately assess its pathological modifying effects, and confirm long-term safety and efficacy in larger-scale clinical trials.

c-Abl inhibitors

c-Abl, a critical non-receptor tyrosine kinase, functions as a key sensor of cellular stress and is widely expressed throughout the brain, playing an indispensable role in neuronal development and survival. Under normal physiological conditions, c-Abl remains in a relatively quiescent state; however, it becomes activated during neurodegeneration [157]. Elevated levels of activated c-Abl have been observed in brain tissues of patients with PD, suggesting that c-Abl inhibitors may hold therapeutic potential for PD. Nilotinib, an oral c-Abl inhibitor initially approved by the FDA for the treatment of chronic myeloid leukemia (CML), has recently been investigated for PD. Studies indicate that nilotinib exhibits favorable safety and tolerability, degrades α -syn, prevents parkin

inactivation, and inhibits c-Abl, thereby potentially mitigating neurodegeneration [158, 159]. Despite its acceptable safety profile, nilotinib demonstrated limited clinical benefit in PD patients, likely due to its poor brain penetration and insufficient cerebrospinal fluid concentrations to achieve meaningful c-Abl inhibition.

Building on the lessons learned from nilotinib's shortcomings, new c-Abl inhibitors with improved central nervous system penetration have been developed. Research indicates that vodobatinib and IKT-148009, two next-generation c-Abl inhibitors, exhibit superior brain penetration and more potent inhibition of cerebral c-Abl compared to nilotinib, showing considerable promise for PD treatment [160–162]. Both compounds are currently in Phase II clinical trials, offering the potential to overcome the therapeutic limitations observed with nilotinib. The field of c-Abl inhibition has thus progressed from the clinical setbacks of nilotinib towards a new generation of high-potency agents. Future efforts must focus on core translational objectives, including direct comparative efficacy analyzes and accelerated biomarker development, to overcome existing bottlenecks and successfully advance this therapeutic class from clinical trials to practical clinical application.

GLP-1 receptor agonists

Glucagon-like peptide 1 (GLP-1) is a peptide hormone and growth factor that can cross the blood–brain barrier and is involved in the regulation of various physiological functions. Its neuroprotective effects have been demonstrated in preclinical studies. Glucagon-like peptide 1 receptor agonists (GLP-1 RAs) have been widely used clinically for the treatment of type 2 diabetes [163].

Recent studies have shown that GLP-1 RAs can improve synaptic transmission in the striatum and protect neurons from PD-related stressors [164], suggesting the potential therapeutic role of GLP-1 in neurodegenerative diseases. Research indicates that the GLP-1 receptor agonist exenatide (Exendin-4) exhibits significant anti-inflammatory effects in neurodegenerative diseases, can clear pathological α -syn through autophagy, and alleviates motor deficits induced by PD [165, 166]. Compared with several other GLP-1 receptor agonists, exenatide demonstrates stronger penetration of the blood–brain barrier and a higher proportion entering the brain, and its clinical trials are more extensive [167]. Currently, phase III clinical trials on the efficacy of exenatide for PD treatment are underway. Other GLP-1 receptor agonists with potential for treating neurodegenerative diseases, such as liraglutide, semaglutide, and lixisenatide, are in phase II clinical trials.

Although GLP-1 RAs represent a promising candidate direction for PD treatment with an optimistic outlook, they still face research bottlenecks such as unclear mechanisms underlying individual differences in efficacy, challenges in medication adherence, and a lack of long-term safety data. Their successful translation ultimately depends on in-depth analysis of their mechanisms of action, the establishment of precise treatment strategies, and comprehensive evaluation of efficacy, safety, and cost-effectiveness.

Statins

Statins are currently widely used in clinical practice as lipid-lowering agents and drugs for the treatment of cardiovascular diseases. Recent studies have shown that lipophilic statins may have certain therapeutic effects on neurodegenerative diseases, potentially exerting their benefits through various mechanisms such as anti-inflammatory and antioxidant actions. Statins are divided into lipophilic and hydrophilic categories. Existing research indicates that long-term use of lipophilic statins can reduce the incidence of PD [168], likely by inhibiting microglial activation and decreasing the accumulation of inflammatory mediators and α -syn, thereby attenuating neuroinflammation [169].

A recent study on lovastatin for PD treatment demonstrated that lovastatin has stronger blood–brain barrier

penetration compared to other statins, shows potential therapeutic effects on motor symptoms in early-stage PD patients, and exhibits good tolerability [170]. Additionally, simvastatin is also highly lipophilic and can cross the blood–brain barrier. Studies in PD animal models have shown that simvastatin exerts significant antioxidant and neuroprotective effects against MPP⁺-induced neurotoxicity [171]. However, a recent clinical trial involving patients with moderate PD indicated that simvastatin failed to demonstrate significant therapeutic efficacy, which contrasts sharply with the optimistic results from prior observational studies and animal experiments [172]. This inconsistency in efficacy assessment may stem from differences in disease stage, statin type, and treatment timing, highlighting the uncertainty regarding its clinical benefits. Therefore, larger-scale studies are needed to explore the therapeutic role of statins in PD. Currently, both simvastatin and lovastatin are in phase II clinical trials, and their results will provide higher-level evidence for clarifying the true value of statins in the treatment of PD.

Iron chelators

Iron plays a crucial role in the human body; however, studies have revealed that iron levels are significantly elevated in the substantia nigra of PD patients. Excessive iron accumulation can damage the nigrostriatal dopaminergic system, induce OS, and promote the aggregation of misfolded α -syn, thereby accelerating PD progression [173]. Growing evidence suggests that iron chelators may have therapeutic potential in PD. Initially used to treat systemic iron overload, iron chelators have subsequently been found to exert neuroprotective effects and cross the blood–brain barrier [174, 175].

In PD treatment, iron chelators primarily act by reducing abnormal cerebral iron deposition. Their core mechanisms include attenuating iron-catalyzed OS and lipid peroxidation, inhibiting ferroptosis—an iron-dependent form of cell death—and disrupting the toxic coupling between iron and α -syn, thereby mitigating its pathological aggregation [26]. A prior clinical study demonstrated that iron chelation reduced iron deposition in the substantia nigra and improved motor symptoms in PD patients, indicating favorable outcomes [176].

However, a recently published phase II clinical trial using the iron chelator deferiprone reported that deferiprone was ineffective in newly diagnosed PD patients who had not previously received dopaminergic therapy and even worsened both motor and non-motor symptoms [177]. This trial had certain limitations, and the efficacy of deferiprone in patients already on dopaminergic medications requires further validation. Notably, the contradiction between the long-term biosafety profile

and the clinical translational potential of iron chelators cannot be overlooked. Animal data and clinical studies indicate that prolonged use may lead to biosafety concerns such as hematological toxicity and depletion of essential trace elements [176, 178].

Therefore, before considering iron chelators as disease-modifying therapies for neurodegenerative disorders requiring lifelong treatment, longer-term clinical studies are necessary to establish systematic safety monitoring protocols and to carefully weigh the potential benefits of reducing cerebral iron load against the risks associated with chronic administration. Moving forward, efforts should focus on developing novel iron chelators with higher brain selectivity and improved safety profiles to achieve an optimal balance between therapeutic efficacy and risk management.

Other investigational agents

β -Glucocerebrosidase (GCase) is a lysosomal enzyme encoded by the GBA1 gene. Mutations in GBA1 represent one of the most common genetic risk factors for PD, leading to significantly reduced GCase protein activity in cholinergic neurons and impairing α -syn metabolism, thereby elevating α -synuclein levels in these neurons [179]. Ambroxol, a cough suppressant widely used in clinical practice, has recently been identified as a molecular chaperone that stabilizes and enhances β -glucocerebrosidase activity, offering a unique potential strategy for disease-modifying therapy in PD. The first clinical trial of ambroxol in PD demonstrated its ability to cross into the cerebrospinal fluid and increase glucocerebrosidase protein levels regardless of the presence of GBA1 mutations [180]. Studies have further confirmed that ambroxol can elevate GCase enzymatic activity and reduce α -synuclein levels [181]. It may modulate α -synuclein levels through multiple mechanisms, indicating that the GCase pathway represents a potential therapeutic target for PD. Currently, a phase II clinical trial of ambroxol for PD is underway. Although early results are promising, they must be interpreted with balance, as future findings may reveal limitations in its efficacy. From a translational perspective, ambroxol offers notable advantages due to its established safety profile and low cost, which could facilitate its rapid repurposing for clinical use.

Terazosin (TZ) is an α 1-adrenergic receptor antagonist primarily used clinically for treating benign prostatic hyperplasia. However, recent studies suggest its potential therapeutic value in PD. Given the impaired mitochondrial function and reduced ATP levels in PD patients, TZ enhances the activity of phosphoglycerate kinase 1 (PGK1), thereby stimulating glycolysis and increasing cellular ATP levels [182]. Research in

PD cellular and animal models treated with MPTP and 6-OHDA has shown that TZ can restore deficient pyruvate and cerebral dopamine levels, elevate in vivo ATP concentrations, alleviate motor symptoms, and consequently suppress or slow neurodegeneration in PD animal models. Furthermore, significant accumulation of TZ in the brain following administration indicates its ability to cross the blood–brain barrier, highlighting its considerable promise for PD treatment [183]. Additional evidence suggests that TZ use is associated with a reduced risk of developing PD, indicating its potential neuroprotective effects [184, 185]. Another study on TZ and cognitive impairment in PD demonstrated that, compared with tamsulosin—another α 1-adrenergic receptor antagonist that does not promote glycolysis—treatment with glycolysis-enhancing TZ was associated with a lower risk of dementia in PD patients [186]. The repurposing of TZ offers a novel perspective for PD treatment; however, its precise neuroprotective mechanisms remain incompletely understood. Future rigorous clinical trials are required to clarify its efficacy specificity, address long-term safety concerns in the PD population, and evaluate its potential cost-effectiveness advantages.

Recent studies suggest that ginsenoside Rg1 possesses therapeutic potential for neurodegenerative diseases. In an LPS-induced mouse model of PD, administration of Rg1 was shown to reduce the activation of microglia and astrocytes, decrease levels of pro-inflammatory cytokines such as TNF- α , IL-1 β , and IL-6, elevate anti-inflammatory factors, and mitigate the loss of nigral dopaminergic neurons [187, 188]. These findings indicate that Rg1 can attenuate inflammatory responses in LPS-induced PD mice, thereby exerting neuroprotective effects and ameliorating PD-related symptoms. Other studies have demonstrated that treatment with ginsenoside Rg1 in animal models of PD effectively reduces the loss of dopaminergic neurons in the substantia nigra and striatum and alleviates motor deficits [189]. Its protective effects against neurodegenerative diseases are likely mediated through multiple mechanisms, highlighting its considerable promise for future PD therapy. Existing animal data support the favorable long-term safety profile of Rg1, with no significant systemic toxicity, tissue damage, or behavioral abnormalities observed following prolonged administration [190]. Subsequent research should focus on conducting more comprehensive long-term safety evaluations to provide a solid foundation for its eventual clinical translation.

Melatonin is a hormone primarily secreted by the pineal gland, with broad physiological functions in the body. It is mainly involved in regulating circadian rhythms and also possesses anti-inflammatory, antioxidant, and

anti-apoptotic properties. Due to its ability to cross the blood–brain barrier, along with its minimal side effects and favorable tolerability, melatonin holds considerable potential for treating sleep disturbances in PD patients [191]. Preclinical studies have demonstrated that melatonin effectively mitigates lipid peroxidation, improves mitochondrial dysfunction, suppresses neuroinflammation, and reduces the loss of nigrostriatal dopaminergic neurons. Its mechanisms involve direct scavenging of free radicals, activation of the Nrf2 pathway to enhance cellular antioxidant defenses, and upregulation of SIRT3 to improve mitochondrial function and biogenesis [192]. Additionally, melatonin inhibits the activation of the NLRP3 inflammasome in microglia, modulates the balance between pro-inflammatory and anti-inflammatory cytokines, and directly interferes with the abnormal aggregation and fibrillation of α -synuclein [193]. In both in vitro and in vivo models of PD, melatonin has been shown to decrease the expression of pro-inflammatory cytokines such as IL-1 β and TNF- α while elevating levels of anti-inflammatory factors. Furthermore, melatonin suppresses MPP⁺-induced activation of the NLRP3 inflammasome, thereby preventing neuronal degeneration and microglial activation, and ameliorates motor symptoms in PD mice [194, 195]. Notably, no significant adverse reactions have been reported in patients following melatonin administration, indicating its relative safety and efficacy [196]. Although current research suggests that melatonin may exert multi-target neuroprotective mechanisms beyond sleep improvement, its precise mechanisms of action in PD remain incompletely understood. Future studies should aim to define its optimal therapeutic time window and design biomarker- and pathology-related clinical endpoints to validate its disease-modifying potential.

Advanced non-pharmacological interventions

Currently, beyond pharmacological treatments, deep brain stimulation (DBS) has become an effective surgical modality for addressing motor complications in PD. This therapy involves implanting electrodes connected to a pulse generator to deliver electrical signals to specific brain targets, thereby modulating neural circuits [197, 198]. Commonly used clinical targets include the subthalamic nucleus (STN) and the globus pallidus internus (GPi), each with distinct therapeutic emphases. STN-DBS demonstrates significant efficacy in improving tremor, rigidity, and bradykinesia, can mildly alleviate anxiety and depression, and enables a substantial reduction in patients' dopaminergic medication requirements [199–201], making it the most widely applied approach in clinical practice. GPi-DBS is particularly effective for medication-induced dyskinesias,

effectively ameliorating dystonia and pain, and allows patients to continue dopaminergic medication without an increased risk of dyskinesias [202]. However, DBS is also associated with potential side effects. GPi-DBS may exacerbate dysarthria [203]; STN-DBS, besides the risk of dysarthria, may also cause eyelid opening apraxia [204, 205], and some studies suggest a potential increase in apathy [206]. It is noteworthy that DBS is not suitable for all patients, as its efficacy and safety are highly dependent on precise target localization, individualized parameter programming, and meticulous postoperative management. Therefore, stringent patient selection, surgery performed by an experienced multidisciplinary team, and long-term follow-up are crucial for maximizing therapeutic benefit.

Immunotherapies targeting α -syn aggregates offer new hope for disease-modifying treatment in PD. The Phase II PASADENA study of prasinezumab, a humanized monoclonal antibody targeting the C-terminus, did not meet its primary endpoint [207]. However, exploratory analyses revealed it could significantly slow the progression of motor symptoms, an effect particularly pronounced in rapid-progressor subgroups, for example, users of monoamine oxidase B inhibitors, patients with diffuse malignant phenotypes [208]. Subsequent research further suggested that its effect on slowing motor and daily functional decline might be sustained long-term [209]. Although its safety profile was generally manageable, infusion reactions were relatively common [210]. These findings indicate that prasinezumab may possess disease-modifying potential in rapidly progressing PD, but its confirmatory efficacy and applicability to broader populations await validation in further clinical trials.

Stem cell therapy, as a highly promising disease-modifying strategy, has achieved key clinical progress in recent years [211]. A landmark Phase I/II clinical trial has been reported. The study involved transplantation of allogeneic iPS-cell-derived dopaminergic progenitors into the putamen of PD patients. Over 24 months of follow-up, the procedure demonstrated a favorable safety and tolerability profile, with no tumor formation or graft-induced dyskinesias observed [212]. ¹⁸F-DOPA PET imaging confirmed graft survival and dopamine synthesis, with superior effects observed in the high-dose group. Clinical assessments showed improvement in motor symptoms in the majority of patients [213]. This study provides the first evidence for the feasibility of using an off-the-shelf iPS cell product for cell replacement therapy in PD, laying a solid foundation for developing standardized, scalable regenerative medicine therapies and representing a significant step towards curing PD.

Gene-editing therapies aim to correct the genetic defects or pathological processes underlying PD at their

source, with the goal of directly intervening in the disease’s genetic or molecular etiology, protecting neurons, and halting or delaying disease progression [214]. Research has found that using CRISPR-Cas9 to edit SNCA and reduce α -syn aggregation can improve motor function in mouse models. Additionally, editing the LRRK2 gene has been shown to partially restore dopaminergic neuron function [215, 216]. Gene-editing therapies face future challenges including delivery, safety, and disease heterogeneity, necessitating multidisciplinary strategies to optimize the therapeutic window.

Among existing non-pharmacological treatments, DBS is the current therapy for managing motor complications in mid-to-late-stage patients. Immunotherapy represents an emerging disease treatment strategy in the clinical validation phase, aiming for early intervention to slow pathological progression. Stem cell therapy is a novel approach moving towards the clinic, aimed at repairing damaged neural networks. Gene editing represents a future hope, particularly for genetic forms of PD, aiming to eliminate the disease cause at its origin. They correspond to different stages of PD disease management and different levels of need (Table 3).

Currently, pharmacological treatment remains the cornerstone throughout the disease course for controlling core motor symptoms. When medications lead to long-term syndrome complications, DBS intervenes as the most mature non-pharmacological means for effective symptomatic relief. Simultaneously, rehabilitative therapy is an essential component of long-term support. In summary, the treatment of PD has entered an era of integrated management characterized by the synergy between pharmacological and non-pharmacological approaches, equal emphasis on symptomatic and causative treatment, and consideration of both present and future needs. In summary, the evolution of Parkinson’s disease treatment can be summarized as follows: a shift from purely symptomatic control to multi-targeted

synergistic effects, and from a predominantly drug-based management model to a multi-modal intervention approach (Fig. 3).

Conclusion

The pathogenesis of PD originates from complex interactions between genetic susceptibility and environmental factors, ultimately converging into four core pathological events: α -syn misfolding, mitochondrial dysfunction, OS, and neuroinflammation. These mechanisms do not operate in isolation but rather form a self-amplifying vicious cycle: mitochondrial damage exacerbates OS and promotes pathological α -syn aggregation; abnormal α -syn in turn impairs mitochondrial function and triggers neuroinflammation; persistent inflammatory responses further aggravate oxidative damage and neuronal loss. Notably, this pathological network is no longer confined to the central nervous system but propagates and amplifies systemically through pathways such as the gut-brain and spleen-brain axes, providing a new perspective for understanding PD as a systemic disorder.

Confronted with the substantial global increase in PD prevalence and the persisting unmet clinical needs in managing both motor and non-motor symptoms, the development of multi-target therapeutic strategies is critically important. The core objective of these strategies is to address the complex, multi-system pathological mechanisms of PD, aiming to achieve superior overall efficacy compared to single-target therapies through synergistic interventions at different pathological nodes. Therefore, in pharmacological management, it is essential not only to employ combination therapies to enhance dopaminergic signaling across multiple pathways and manage motor complications, but also to actively develop novel agents with multi-target profiles and explore treatments targeting non-dopaminergic systems that possess disease-modifying potential. A higher-level multi-target approach involves the deep integration and personalized

Table 3 Non-pharmacological interventions for Parkinson’s disease

Intervention Type	Mechanism of Action	Clinical Stage	Efficacy
Deep brain stimulation precursor drugs	Electrical stimulation of STN/GPI modulating neural circuits	Widely used in clinical practice	Improves tremor, rigidity, bradykinesia; reduces medication requirements; effectively controls dyskinesia
Immunotherapy	Monoclonal antibody targeting C-terminus of α -synuclein	Phase II clinical trial	Significantly slows motor symptom progression in rapid-progressor subgroups
Stem cell therapy	Transplantation of iPS cell-derived dopaminergic progenitors	Phase I/II clinical trial	Favorable safety and tolerability at 24 months; motor symptoms improved in most patients
Gene editing therapy	CRISPR editing of SNCA/LRRK2 genes	Preclinical research	Improves motor function in animal models

This table summarizes the key non-pharmacological interventions for Parkinson’s disease, including deep brain stimulation, immunotherapy, stem cell therapy, and gene editing therapy. For each intervention type, the mechanism of action, current clinical stage, and primary efficacy findings are presented

STN Subthalamic nucleus, GPI Globus pallidus internus, iPS induced pluripotent stem cell

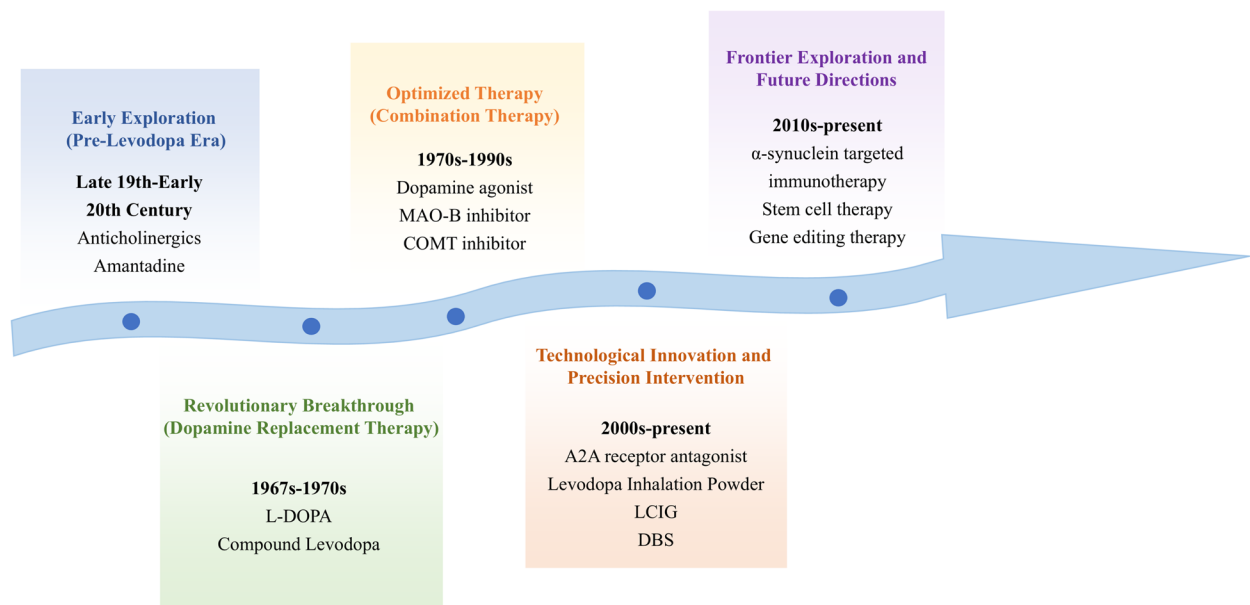


Fig. 3 Milestones in the development of Parkinson's disease therapies. This figure outlines key milestones in the treatment of Parkinson's disease, whose evolution can be summarized in five phases. The early phase relied on symptomatic management using anticholinergic agents. The 1960s-70 s saw the introduction of L-DOPA and its combination formulations, establishing dopamine replacement as the cornerstone therapy. Subsequently, adjuvant agents such as dopamine agonists, MAO-B inhibitors, and COMT inhibitors, together with continuous infusion strategies, were developed to optimize efficacy and manage motor complications. In the twenty-first century, treatment options expanded further to include novel drug-delivery systems, deep brain stimulation (DBS) surgery, and the first non-dopaminergic agent (adenosine A2A receptor antagonist). Current frontiers are shifting toward disease-modifying strategies, including α -synuclein-targeted immunotherapy, stem cell-based approaches, and gene-editing therapies, marking a transition from symptomatic control to interventions aimed at slowing or repairing the underlying disease process

application of pharmacological and non-pharmacological interventions. For instance, combining optimized post-deep brain stimulation (DBS) medication regimens with targeted rehabilitation can achieve synergy between pharmacological modulation and neural circuit remodeling. This multidimensional strategy aims to concurrently ameliorate motor and non-motor symptoms, balance short-term symptomatic control with long-term disease modification, thereby delaying disease progression and comprehensively improving patients' quality of life. In the future, as the understanding of pathological mechanisms deepens, multi-target therapies are anticipated to evolve toward more precise, dynamic, and individualized paradigms.

To achieve effective treatment for PD, future research must achieve breakthroughs in several key areas. First is the realization of early diagnosis and precise classification of the disease, providing a basis for subsequent targeted, individualized treatment. Second, given the complexity of PD pathogenesis, developing multi-target synergistic therapies capable of simultaneously intervening in multiple critical pathways is particularly crucial. This strategy aims to achieve superior disease control

through combined effects. Concurrently, elucidating the interaction mechanisms between the brain and peripheral organs such as the gut and spleen may not only reveal novel pathways for pathological propagation but also uncover a series of entirely new intervention targets. Finally, accelerating the transition of promising frontier strategies—such as gene therapy and cell therapy—from laboratory research to clinical validation and application is a critical step toward achieving the ultimate goals of neural repair and disease modification. Through sustained exploration and systematic integration across these domains, PD treatment paradigms are poised to make a fundamental leap from symptom management toward neural repair and functional restoration.

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Authors' contributions

Shanshan Zhang wrote the manuscript and drew the figures and tables. Tingyu Wang participated in the revision. Ye Peng, Qianchen Wang, and Zhao Zhang contributed to the study design, refined the manuscript, and provided detailed feedback on its content and presentation. Shifeng Chu, Hefei Huang, and Naihong Chen reviewed the manuscript and provided critical comments. All authors read and approved the final manuscript.

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Data availability

Not applicable.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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